Md. Farhan Shahriyar

Quantum Machine Learning · Cybersecurity · Cryptography

Dinajpur, Bangladesh | mfshahriyar.github.io | Google Scholar | ResearchGate | GitHub

RESEARCH INTERESTS

Quantum machine learning (quantum feature maps, quantum kernels, hybrid quantum-classical models); quantum key distribution and cryptography; threat and phishing detection; feature selection and algorithmic efficiency; privacy-preserving learning; efficient and reliable AI for security.

EDUCATION

M.Sc. in Computer Science and Engineering, Hajee Mohammad Danesh Science and Technology University (HSTU), Dinajpur — 2025 – Present

Thesis: *NirapodQSeal: A Hybrid Quantum–Classical Framework for Secure File Transmission and Concealment via Entanglement and Hash Integrity*
Supervisor: Dr. Md. Abdulla Al Mamun, Professor and Chairman, Department of CSE, HSTU
Funding: NST Fellowship 2025–2026, Ministry of Science and Technology, Bangladesh

PUBLICATIONS

[1] **M. F. Shahriyar**, M. A. Al Mamun. "Quantum Key Distribution With Steganography: A Systematic Review of Hybrid Security Frameworks for Postquantum Cryptography." *Quantum Engineering*, 2026. DOI: 10.1155/que2/7042681

[2] **M. F. Shahriyar**, G. Tanbhir, A. M. R. Chy. "Quantum Machine Learning for Image Classification: A Hybrid Model of Residual Network with Quantum Support Vector Machine." *IEEE NCIM*, 2025. DOI: 10.1109/NCIM65934.2025.11160179

[3] **M. F. Shahriyar**, G. Tanbhir, A. M. R. Chy, M. A. A. A. Tanzin, M. J. Mashrafi. "PhishVQC: Optimizing Phishing URL Detection with Correlation Based Feature Selection and Variational Quantum Classifier." *arXiv:2503.01799*, 2025.

Project report: "NirapodSobzi: A Vegetable Disease Detection Application" (project proposal, CSE 531, HSTU), 2026. DOI: 10.13140/RG.2.2.27319.07845/3

EXPERIENCE

Team Lead, ZogBiyog — Present

Researcher, Department of CSE, HSTU (quantum ML, QKD and AI for cybersecurity) — 2025 – Present

Programming Team Lead, WUB dx0 Programming Team — 01/2021 – 12/2023

Activity Coordinator, WUB Computer Society — 01/2020 – 12/2023

Mentorship: research partner and mentor to Abdullah Md Raihan Chy (co-author), from early research work through Ph.D. applications; he received Ph.D. offers from seven universities.

HONORS AND AWARDS

NST Fellowship 2025–2026, Ministry of Science and Technology, Bangladesh (Physical Science) — 2026

ICPC 2023 Regional Round, Honorable Mention — 2023

National Merit Scholarship (SSC), Ministry of Education, Bangladesh — 2015 – 2017

CERTIFICATIONS

Google Cybersecurity Professional Certificate (8 courses), Google / Coursera — Nov 2023

Master Git and GitHub: Beginner to Expert, Udemy — Sep 2024

SKILLS

Languages: Python, SQL, Bash, C, C++
Quantum & ML: Qiskit, PennyLane, PyTorch, TensorFlow, Scikit-learn
Frameworks: Django
Tools: Git, GitHub, Linux, LaTeX, Jupyter, VS Code, Zotero, PowerShell, Microsoft Office